

Computer Science & DA

Calculus and Optimization



DPP 02 Discussion Notes

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[MCQ]



#Q.

A 2m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec, then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is:

A

$$25\sqrt{3}$$

B

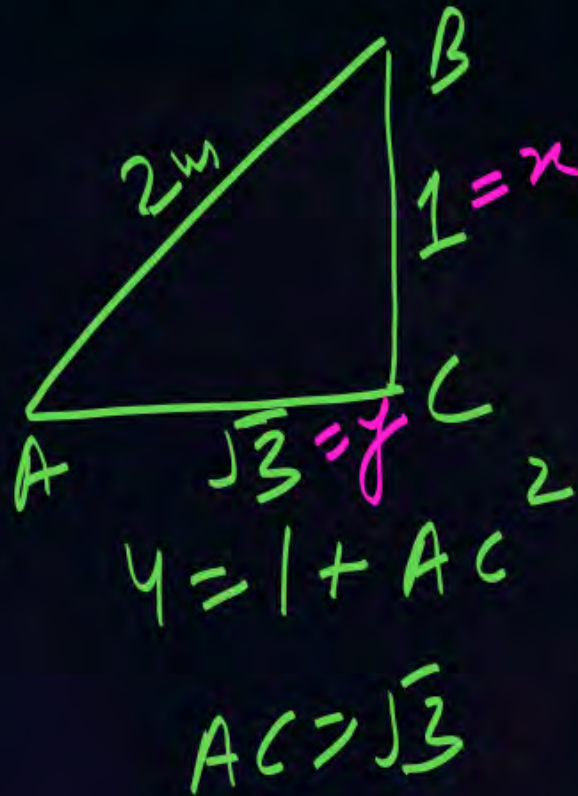
$$25$$

C

$$\frac{25}{\sqrt{3}}$$

D

$$\frac{25}{3}$$



$$x^2 + y^2 = 4 \quad \text{--- (1)}$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$(1)(-25) + \sqrt{3} \left(\frac{dy}{dt} \right) = 0$$

$$\frac{dy}{dt} = \frac{25}{\sqrt{3}}$$

[MCQ]



#Q. The percentage error in calculating the volume of a cubical box if an error of 1% is made in measuring the length of edges of the cube is :

A 1%

B 2%

C 3%

D 6%

$$\begin{aligned}\text{Absolute Error} &= dn \\ \text{Relative " } &= \frac{dn}{n} \\ \% \quad " &= \frac{dn}{n} \times 100\end{aligned}$$

$$\begin{aligned}V &= n^3 \quad \text{--- (1)} \\ dV &= 3n^2 dn \\ \frac{dV}{V} &= \frac{3n^2}{V} dn \\ \frac{dV}{V} &= \frac{3}{n} dn \\ \frac{dV}{V} \times 100 &= 3 \left(\frac{dn}{n} \times 100 \right) = 3 \times 1\% \\ &= 3\%\end{aligned}$$

#Q. Find the tangent line to $f(x) = 7^x + 4e^x$ at $x = 0$.

$$\text{At } x=0, y = f(0) = 7^0 + 4e^0 = 5$$

$$\text{So point of tangent} = (x, y) = (0, 5) = P$$

$$\text{Slope of tangent } m = \left(\frac{dy}{dx} \right)_P = \left(7^x \cdot \log_e 7 + 4e^x \right)_{P(0,5)} = \log 7 + 4$$

$$\text{eqn of tangent } y - y_1 = m(x - x_1)$$

$$y - 5 = (\log 7 + 4)(x - 0)$$

Ans

#Q. Differentiate $f(t) = \frac{1 + 5t}{\ln(t)}$

$$f'(t) = \frac{\ln t (0+5) - (1+5t) \left(\frac{1}{t}\right)}{(1+\ln t)^2} = \frac{5\ln t - \frac{1}{t} + 5}{(1+\ln t)^2}$$

[NAT]



#Q. Find $\frac{\partial w}{\partial x}$ for the following-

$$w = \cos(x^2 + 2y) - e^{4x - z^4 y} + (y^3)$$

$$\begin{aligned}\frac{\partial w}{\partial x} &= -\sin(x^2 + 2y)(2x + 0) - e^{4x - z^4 y}(4 - 0) + 0 \\ &= -2x \sin(x^2 + 2y) - 4e^{4x - z^4 y} \quad //\end{aligned}$$

#Q. Find all the 1st order partial derivatives of the following function.

$$f(x, y, z) = 4x^3y^2 - e^z y^4 + \frac{z^3}{x^2} + 4y - x^{16}$$

$$\frac{\partial f}{\partial x} = (12x^2)y^2 - 0 + z^3\left(-\frac{2}{x^3}\right) + 0 - 16x^{15} \quad ,,$$

$$\frac{\partial f}{\partial y} = 4x^3(2y) - e^z(4y^3) + 0 + 4 - 0 \quad ,,$$

$$\frac{\partial f}{\partial z} = 0 - y^4(e^z) + \frac{3z^2}{x^2} + 0 - 0 \quad ,,$$

#Q. Prove that

$$f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$$

is homogeneous; what is the degree? Verify Euler's Theorem for f .

By observation f is n.f of degree $3 = n$

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f$$

$$\boxed{x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 3f} \quad (1)$$

$$LHS = x(3x^2 - 4xy + 3y^2) + y(-2x^2 + 6xy + 3y^2)$$

$$= \dots = 3f = RHS$$

#Q. If $v(x, y) = \log \left(\frac{x^2 + y^2}{x + y} \right)$, prove that $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1$

Sol: $\because v(\lambda x, \lambda y) \neq \lambda^n v(x, y)$ So v is H-Homog

Let $\boxed{e^v = u} = \frac{x^2 + y^2}{x + y} \quad \text{--- (1)}$

$\therefore u(\lambda x, \lambda y) = \lambda^1 u(x, y)$ So u is Homog f^n of $n=1$

So Applying E.T for u is $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n u$

$$x \frac{\partial (e^v)}{\partial x} + y \frac{\partial (e^v)}{\partial y} = 1 \cdot e^v$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1 \cdot \frac{e^v}{e^v} = 1$$

$$\log_e a = b$$

$$\boxed{a = e^b}$$

#Q. Prove that

$$g(x, y) = x \log(y/x)$$

is homogenous, what is the degree? Verify Euler's theorem for g .

$$\rightarrow \underline{g(\lambda x, \lambda y)} = (\lambda x) \log\left(\frac{\lambda y}{\lambda x}\right) = \lambda \left[x \log \frac{y}{x} \right] = \lambda [g(x, y)] \quad \text{As } g \text{ is h.f.d } (n=1)$$

$$\text{E.Th is } x \frac{\partial g}{\partial x} + y \frac{\partial g}{\partial y} = 1 \cdot g \quad \text{--- (1)}$$

$$\begin{aligned} \frac{\partial g}{\partial x} &= x \left(\frac{1}{y/x} \cdot \frac{-y}{x^2} \right) + \log\left(\frac{y}{x}\right) [1] \\ &= -1 + \log\left(\frac{y}{x}\right) \end{aligned}$$

$$\begin{aligned} \frac{\partial g}{\partial y} &= x \left[\frac{1}{y/x} \cdot \frac{1}{x} \right] = \frac{x}{y} \\ \text{LHS of (1)} &= x \left[-1 + \log\left(\frac{y}{x}\right) \right] + y \left[\frac{x}{y} \right] \\ &= x \log \frac{y}{x} = 1 \cdot g(x, y) \\ &= \text{RHS} \end{aligned}$$

#Q. The jacobian of p, q, r w. r. t x, y, z given, $p = x + y + z$, $q = y + z$, $r = z$ is _____.

$$J = \frac{\partial(p, q, r)}{\partial(x, y, z)} = \begin{vmatrix} p_x & p_y & p_z \\ q_x & q_y & q_z \\ r_x & r_y & r_z \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad \underline{\underline{\text{Ans}}}$$

[MCQ]



#Q.

Given $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$ then the value of $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ is ____.

A

4

C

0

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

$$= \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix}$$

B

-4

D

1

$$J = \frac{1}{x^2 y^2 z^2} \begin{vmatrix} -yz & zx & xy \\ zy & -zx & xy \\ yz & xz & -xy \end{vmatrix}$$

$$R_2 \rightarrow R_2 + R_1, \text{ \& } R_3 \rightarrow R_3 + R_1$$

$$= \frac{1}{x^2 y^2 z^2} \begin{vmatrix} -yz & zx & xy \\ 0 & 0 & 2xy \\ 0 & 2zx & 0 \end{vmatrix}$$

$$= \frac{1}{x^2 y^2 z^2} (-4x^2 yz) = \frac{-4}{yz}$$

[MSQ]



$$D_f : x > 2 \text{ or } (2, \infty)$$

#Q. The function $f(x) = 2\log(x-2) - x^2 + 4x + 1$ increase on the interval

$$f'(x) = \frac{2}{x-2} - 2x + 4 = \frac{2 - 2(x-2)^2}{(x-2)} = \frac{2 - 2[x^2 + 4 - 4x]}{x-2}$$

A (1, 2)

C (5/2, 3)

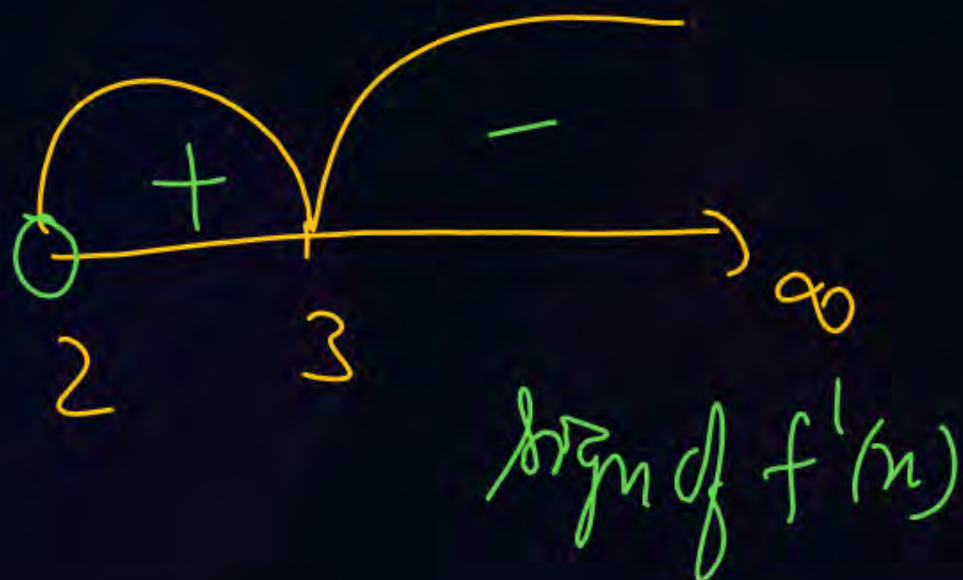
B (2, 3)

D (2, 4)

$$= \frac{2 - 2x^2 - 8 + 8x}{x-2}$$

$$= -2 \left[\frac{x^2 - 4x + 3}{x-2} \right]$$

$$f'(x) = -2 \left[\frac{(x-3)(x-1)}{x-2} \right]$$



T-Points are $x=1, 3, 2$
But Acc to Dom. T-Point $x=3$

#Q.

The function $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is $f'(x) = \frac{(x^2 + 1)(2x) - (x^2 - 1)(2x)}{(x^2 + 1)^2} = \frac{2x^3 + 2x - 2x^3 + 2x}{(x^2 + 1)^2}$

$$f''(x) = \frac{(x^2 + 1)^2(4) - 4x[2(x^2 + 1)(2x)]}{(x^2 + 1)^4}$$

$$f'(x) = \frac{4x}{(x^2 + 1)^2}$$

A

is concave for $x \in \left[-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

B

is convex for $x \in \left[-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

C

has 2 points of inflexion

D

is concave for $\left[\frac{1}{\sqrt{3}}, \infty\right)$

$$= 4 \left[\frac{(x^2 + 1) \{ (x^2 + 1) - x(4x) \}}{(x^2 + 1)^4} \right] = 4 \frac{(1 - 3x^2)}{(x^2 + 1)^3} = \frac{-12 \left(x^2 - \frac{1}{3}\right)}{(x^2 + 1)^3} = \frac{-12 \left(x - \frac{1}{\sqrt{3}}\right) \left(x + \frac{1}{\sqrt{3}}\right)}{(x^2 + 1)^3}$$

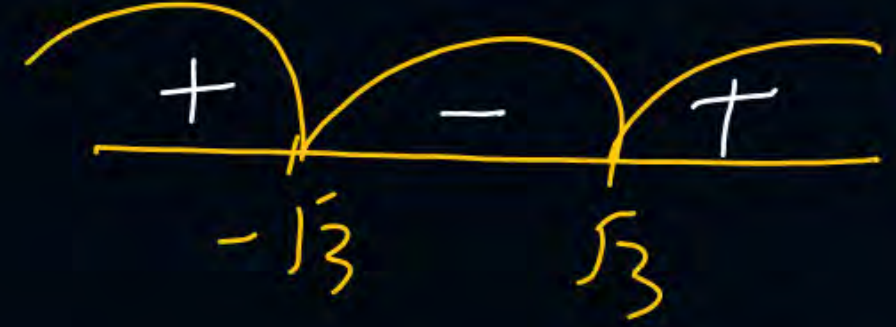


Sign of $f''(x)$

[MCQ]

#Q. Number of the points of inflexion of the function:

$$f(x) = x^4 - 18x^2 + 9$$



$$f'(x) = 4x^3 - 36x$$

$$f''(x) = 12x^2 - 36 = 12(x^2 - 3) = 12(x - \sqrt{3})(x + \sqrt{3})$$

$f''(x)$

A 1

B 2

C 3

D 4

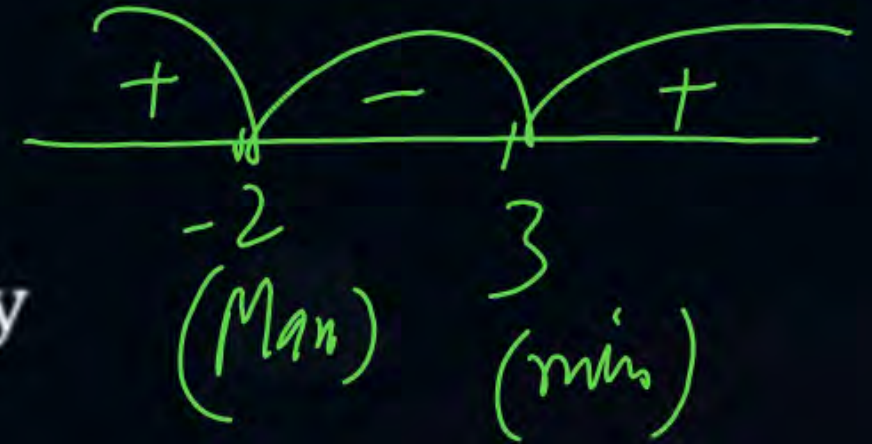
[MCQ]



#Q. The function $f(x) = 2x^3 - 3x^2 - 36x + 2$ has its maxima at

$$f'(x) = 6x^2 - 6x - 36 = 6(x^2 - x - 6) = 6[(x-3)(x+2)]$$

So T. Points are $x = -2, 3$



A ☒ $x = -2$ only

B ☐ $x = 0$ only

C ☐ $x = 3$ only

D ☐ both $x = -2$ and $x = 3$

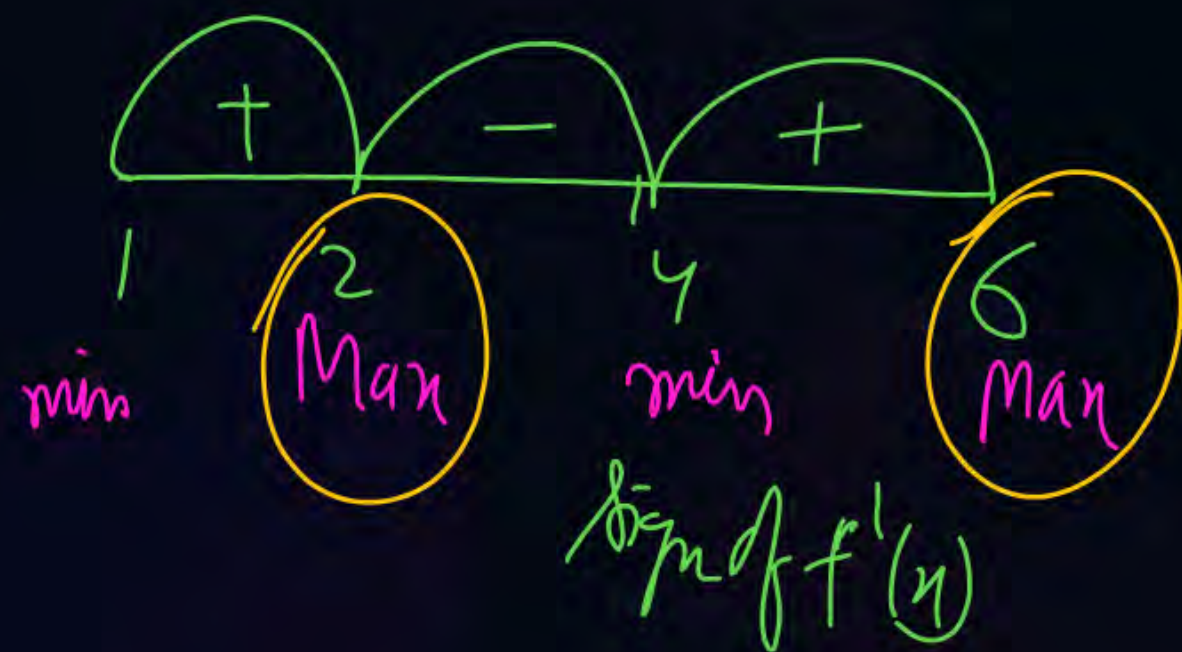
#Q. The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is

$$f'(x) = 3x^2 - 18x + 24 = 3(x^2 - 6x + 8) = 3(x-4)(x-2)$$

∴ Points are $f'(x) = 0 \Rightarrow x = 2, 4$

$$f(2) = 8 - 36 + 48 + 5 = 25$$

$$f(6) = 216 - 324 + 144 + 5 = 41$$



#Q. The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is

$$f'(x) = 3x^2 - 6x - 24$$

$$= 3(x^2 - 2x - 8) = 3(x-4)(x+2)$$



$$\begin{array}{r} 172 \\ 54 \\ \hline \end{array}$$

$$\text{T. Points} \Rightarrow x = -2, 4$$

But Acc to given interval, only T.P

$$f(-3) = -27 - 27 + 72 + 100 = 118 \quad \text{is } x = -2$$

$$f(3) = 27 - 27 - 72 + 100 = 28 \quad \underline{\underline{Ans}}$$

[MCQ]



#Q. Suppose that the amount of money in a bank account after t year is given

by $A(t) = 2000 - 10te^{5-\frac{t^2}{8}} \Rightarrow \frac{dA}{dt} = 0 - 10 \left[e^{5-\frac{t^2}{8}} (1) + t \left\{ e^{5-\frac{t^2}{8}} \left(-\frac{2t}{8}\right) \right\} \right]$

The minimum and maximum amount of money in the account during the first 10 year that it is open occur respectively at:

A ✓ $t = 2, t = 0$

$A(0) = 2000$

B $T = 2, t = 10 = -10e^{5-\frac{t^2}{8}} \left[1 - \frac{t^2}{4} \right]$

C ✗ $T = 0, t = 2$

$A(10) < 2000$

D ✗ $T = 10, t = 0 = \frac{10}{4} e^{5-\frac{t^2}{8}} (t^2 - 4)$



$\frac{dA}{dt} = \frac{10}{4} e^{5-\frac{t^2}{8}} (t-2)(t+2)$
 $\Rightarrow t = 2, (-2) \times$

[MSQ]

#Q. The function: $f(x) = x^2 \ln(3x) + 6$ has:

$$D_f = (0, \infty)$$

$$f'(x) = 0$$

$$x^2 \left(\frac{1}{3x} \cdot 3 \right) + \ln(3x) \cdot (2x) = 0$$

$$x + 2x \ln 3x = 0 \Rightarrow$$

$$x(1 + 2 \ln 3x) = 0 \Rightarrow \begin{cases} x = 0 \\ 1 + 2 \ln 3x = 0 \end{cases}$$

A

No critical point

B

Only one critical point

C

Only one stationary point

D

Infinitely many critical points

$$x = \frac{1}{\sqrt{3e}}$$

$$x = \pm \frac{1}{\sqrt{3e}}$$

$$\ln(3x)^2 = -1$$

$$3x^2 = e^{-1} \Rightarrow x^2 = \frac{1}{3e}$$

[MCQ]

#Q. The function $f(x,y) = x^2y - 3xy + 2y + x$ has

- A** No local extremum
- B** One local minimum but no local maximum
- C** One local maximum but no local minimum
- D** One local minimum and one local maximum

$$f_x = 2xy - 3y + 1$$

$$f_{xx} = 2y$$

$$f_y = x^2 - 3x + 2$$

$$f_{yy} = 0$$

$$f_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (x^2 - 3x + 2) = 2x - 3$$

$$\Delta f = f_{xx}^2 - f_{xy}^2 = 0 - (2x - 3)^2 < 0$$

[MCQ]

#Q. Find the local minima of the function $f(x, y) = 2x^2 + 2xy + 2y^2 - 6x$

$$f_x = 4x + 2y - 6, \quad f_y = 2x + 4y, \quad f_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (2x + 4y) = 2$$

$$f_{xx} = 4, \quad f_{yy} = 4$$

A

(2, 1)

B

(-2, 1)

C

(2, 0)

D

(2, -1)

$$f_x = 0 \quad \& \quad f_y = 0$$

$$\begin{cases} 4x + 2y = 6 \\ 2x + 4y = 0 \end{cases} \Rightarrow x = 2, y = -1$$

$$\Delta = f_{xx}f_{yy} - (f_{xy})^2 = (4)(4) - (2)^2 = 12 > 0$$

#Q. Find the critical points of the function $f(x, y) = y^2 - x^2$ and check for the presence of any saddle point.

$$f_x = 0, f_y = 0$$

$$-2x = 0, 2y = 0$$

$$x = 0, y = 0$$

$P(0, 0)$ is the C. Point.

$$r = f_{xx} = -2$$

$$t = f_{yy} = 2$$

$$s = f_{xy} = 0$$

$$rt - s^2 = () () - ()^2$$

$$= (-2)(2) - (0)^2$$

$$< 0$$

#Q. A jet of an enemy is flying along the curve $y = x^2 + 2$. A soldier is placed at the point (3, 2). What is the nearest distance between the soldier and the jet?

$x =$

$$AP = \sqrt{(x-3)^2 + (y-2)^2}$$

$$AP = \sqrt{(x-3)^2 + (x^2)^2}$$

Let $AP^2 = U = (x-3)^2 + x^4$ — (1)

where $AP_{\min} = \sqrt{U_{\min}} = ?$

$AP_{\min} = \sqrt{5}$ Ans

$$\begin{aligned} \frac{dy}{dx} &= 2(x-3) + 4x^3 \\ &= 4x^3 + 2x - 6 \Rightarrow \frac{dy}{dx} = 0 \text{ will be} \end{aligned}$$

when $x = 1$

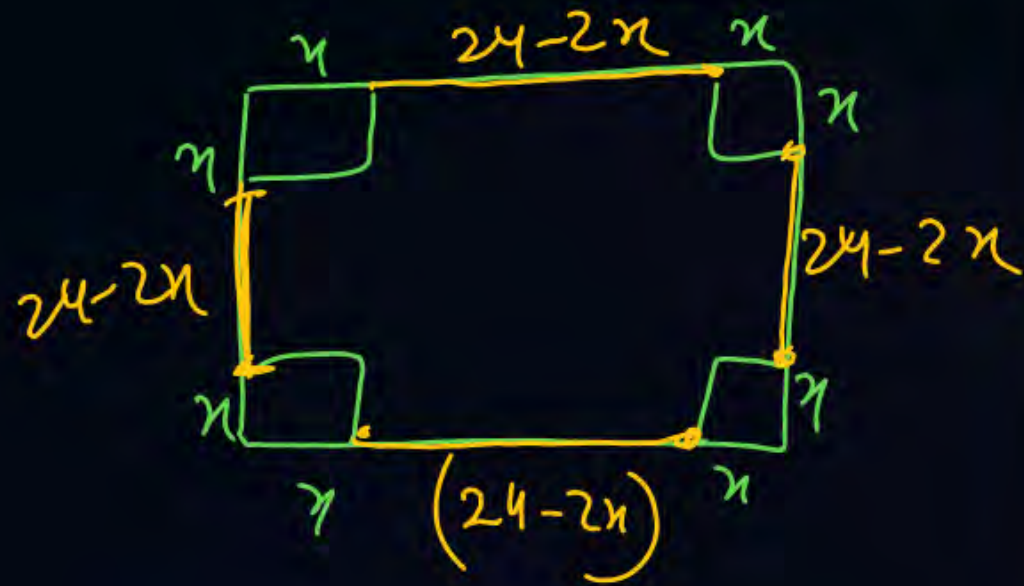
$$\left(\frac{d^2U}{dx^2} \right) = (12x^2 + 2) = +ve$$

$x = 1$

i.e. U will be min at $x = 1$

$$U_{\min} = (1-3)^2 + 1^4 = 5$$

- #Q. A Square piece of tin of side 24 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. What should be the side of the square to be cut off so that the volume of the box is maximum? Also find this maximum volume.



Height = x

$$V = l \times b \times h = (24-2x)(24-2x)x = 16 \times 16 \times 4$$

$$V = (576 + 4x^2 - 96x)x = 4x^3 - 96x^2 + 576x$$

for Max. Volume $\frac{dV}{dx} = 0$

$$12x^2 - 192x + 576 = 0$$

$$12(x^2 - 16x + 48) = 0$$

$$12(x-12)(x-4) = 0$$

$$x = 4 \text{ \& } 12$$

✓ ✗

$$V_{\text{max}} = 4 \times 64 - 96 \times 16 + 576 \times 4$$

$$= 1024$$

#Q. Manufacturer can sell x items at a price of rupees $\left(5 - \frac{x}{100}\right)$ each.

The cost price is Rs. $\left(\frac{x}{5} + 500\right)$ = Total C.P

Find the number of items he should sell to earn maximum profit.

$$P = \text{Total SP} - \text{Total CP}$$

$$= \left(5 - \frac{x}{100}\right) \cdot x - \left(\frac{x}{5} + 500\right)$$

$$P = -\frac{x^2}{100} + 5x - \frac{x}{5} - 500 \quad \text{--- (1)}$$

$$\frac{dP}{dx} = -\frac{x}{50} + 5 - \frac{1}{5} = -\frac{x}{50} + \frac{24}{5} = \frac{-x + 240}{50}$$

$$\frac{d^2P}{dx^2} = -\frac{1}{50} (< 0) \Rightarrow \text{Maxima}$$

$$\frac{dP}{dx} = 0 \Rightarrow x = 240 \quad \underline{\underline{A}}$$

#Q. Evaluate $\int x^2 \sin x \, dx$

$$\int uv \, dx = uv_1 - u'v_2 + u''v_3 - \dots$$

$$\int \underbrace{x^2}_u \cdot \underbrace{\sin x}_v \, dx = x^2(-\cos x) - 2x(-\sin x) + 2(\cos x) - 0(?) + 0 - \dots$$

#Q.

$$\int \frac{dx}{2\sqrt{x}(x+1)} = \frac{1}{2} \int \frac{2t dt}{t \cdot (t^2+1)} = \int \frac{dt}{t^2+1} = \tan^{-1}(t) + C$$

Put $x = t^2$

$$dx = 2t dt$$

$$= \tan^{-1}(\sqrt{x}) + C$$

[NAT]



#Q.

Evaluate $\int \frac{1 \cdot dx}{(x-1)(x-2)(x-3)}$

$$= \int \left(\frac{1/2}{x-1} + \frac{-1}{x-2} + \frac{1/2}{x-3} \right) dx$$

$$\frac{1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

$$= \frac{1}{2} \log(x-1) - 1 \log(x-2) + \frac{1}{2} \log(x-3)$$

$$\checkmark 1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

$$= \log(x-1)^{1/2} + \log(x-3)^{1/2} - \log(x-2)$$

$$\text{Put } (x=1), 1 = A(-1)(-2) \Rightarrow A = 1/2$$

$$\text{Put } (x=2), 1 = B(1)(-1) \Rightarrow B = -1$$

$$\text{Put } (x=3), 1 = C(2)(1) \Rightarrow C = 1/2$$

$$= \log \left[\frac{\sqrt{(x-1)(x-3)}}{x-2} \right] //$$

#Q.

Evaluate

$$I = \int_2^7 \frac{\sqrt{x} dx}{\sqrt{x} + \sqrt{9-x}} \quad \text{--- (1)}$$

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \quad \text{--- (2)}$$

$$(1) + (2) \Rightarrow$$

$$2I = \int_2^7 \left(\frac{\sqrt{x} + \sqrt{9-x}}{\sqrt{x} + \sqrt{9-x}} \right) dx = \int_2^7 (1) dx$$

$$2I = 5$$

$$I = \frac{5}{2}$$

[NAT]



#Q.

Evaluate $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

$$f(x) = \frac{1}{(1+e^x)(1+x^2)} \quad \text{--- (1)}$$

$$\begin{aligned} f(-x) &= \frac{1}{(1+e^{-x})(1+x^2)} \\ &= \frac{e^x}{(1+e^x)(1+x^2)} \quad \text{--- (2)} \end{aligned}$$

$$f(x) + f(-x) = \frac{(1+e^x)}{(1+e^x)(1+x^2)} = \left(\frac{1}{1+x^2} \right)$$

$$I = \int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$$

$$= \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(-x) = f(x) \\ 0, & \text{if } f(-x) = -f(x) \end{cases}$$

$$I = \int_0^{\sqrt{3}} \frac{1}{1+x^2} dx = \left[\tan^{-1}(x) \right]_0^{\sqrt{3}}$$

$$= \tan^{-1}(\sqrt{3}) - \tan^{-1}(0)$$

$$= \frac{\pi}{3} \quad |$$

#Q.

Evaluate $\int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx = \int_{-a}^a \sqrt{\frac{a-x}{a+x}} \times \sqrt{\frac{a-x}{a-x}} dx = \int_{-a}^a \frac{a-x}{\sqrt{a^2-x^2}} dx$

$$I = \int_{-a}^a \frac{a}{\sqrt{a^2-x^2}} dx - \int_{-a}^a \left(\frac{x}{\sqrt{a^2-x^2}} \right) dx$$

$$= a \cdot 2 \int_0^a \frac{1}{\sqrt{a^2-x^2}} dx - 0$$

$$I = 2a \int_0^a \frac{1}{\sqrt{a^2-x^2}} dx = 2a \left(\sin^{-1}\left(\frac{x}{a}\right) \right)_0^a = 2a \left[\sin^{-1}(1) - \sin^{-1} 0 \right] = 2a \left(\frac{\pi}{2} \right) = \pi a$$

#Q.

Evaluate

$$\int_0^{\pi} \frac{x dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{x dx}{1 + \cos^2 x}$$

$$\text{--- (1) } \int_0^{\alpha} f(x) dx = \begin{cases} 2 \int_0^{\alpha/2} f(x) dx, & f(\alpha-x) = f(x) \\ \int_0^{\alpha} f(x) dx, & f(\alpha-x) = -f(x) \end{cases}$$

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$I = \int_0^{\pi} \frac{\pi-x}{1+\cos^2 x} dx = \int_0^{\pi} \frac{\pi}{1+\cos^2 x} dx - I$$

$$2I = \pi \int_0^{\pi} \frac{dx}{1+\cos^2 x} = 2\pi \int_0^{\pi/2} \frac{dx}{1+\cos^2 x}$$

$$I = \pi \int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x + 1} dx$$

$$I = \pi \int_0^{\pi/2} \frac{\sec^2 x}{2 + \tan^2 x} dx$$

$$0, f(\alpha-x) = -f(x)$$

$$\text{Put } \tan x = t$$

$$\sec^2 x dx = dt$$

$$= \pi \int_0^{\infty} \frac{dt}{2+t^2}$$

$$= \pi \left[\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) \right]_0^{\infty} = \frac{\pi}{\sqrt{2}} \left[\frac{\pi}{2} - 0 \right] = \frac{\pi^2}{2\sqrt{2}}$$

[NAT]



#Q.

Evaluate

$$\int_0^{\pi} \frac{dx}{1+2\sin^2 x} = \int_0^{\pi} \frac{dn}{1+2\sin^2 n} = 2 \int_0^{\pi/2} \frac{dn}{1+2\sin^2 n} = 2 \int_0^{\pi/2} \frac{\sec^2 n \, dn}{\sec^2 n + 2\tan^2 n}$$

$$\int_0^{\alpha} f(n) \, dn = \begin{cases} 2 \int_0^{\alpha/2} f(n) \, dn, & \text{if } f(\alpha-n) = f(n) \\ 0, & \text{if } f(\alpha-n) = -f(n) \end{cases}$$

Put $\tan n = t$ $\begin{cases} t=0 \\ t=\infty \end{cases}$
 $\sec^2 n \, dn = dt$

$$= 2 \int_0^{\pi/2} \frac{\sec^2 n \, dn}{1+3\tan^2 n} = \frac{2}{3} \int_0^{\pi/2} \frac{\sec^2 n \, dn}{\left(\frac{1}{\sqrt{3}}\right)^2 + \tan^2 n} = \frac{2}{3} \int_0^{\infty} \frac{dt}{\left(\frac{1}{\sqrt{3}}\right)^2 + t^2} = \frac{2}{3} \frac{1}{(1/\sqrt{3})} \left[\tan^{-1} \left(\frac{t}{1/\sqrt{3}} \right) \right]_0^{\infty}$$

$$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi}{\sqrt{3}}$$

[MCQ]

#Q. Let $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3, x > 0$ Then $18 \int_1^2 \underline{f(x)} dx$ is equal to:

- A** $10 \log 2 - 6$
- B** $10 \log 2 + 6$
- C** $5 \log 2 - 6$
- D** $5 \log 2 - 3$

$$5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3 \quad \text{--- (1)} \Rightarrow 25f(x) + 20f\left(\frac{1}{x}\right) = \frac{5}{x} + 15$$

$$5f\left(\frac{1}{x}\right) + 4f(x) = x + 3$$

$$\text{i.e. } 4f(x) + 5f\left(\frac{1}{x}\right) = x + 3 \quad \text{--- (2)} \Rightarrow 16f(x) + 20f\left(\frac{1}{x}\right) = 4x + 12$$

$$\text{(1)} \times 5 - \text{(2)} \times 4 \Rightarrow$$

$$9f(x) = \frac{5}{x} - 4x + 3$$

$$f(x) = \frac{5}{9x} - \frac{4x}{9} + \frac{3}{9}$$

$$\begin{aligned}
 I &= 18 \int_1^2 f(x) dx = 18 \int_1^2 \frac{1}{9} \left(\frac{5}{x} - 4x + 3 \right) dx = 2 \int_1^2 \left(\frac{5}{x} - 4x + 3 \right) dx \\
 &= 2 \left[5 \log x - 4x^2 + 3x \right]_1^2 \\
 &= 2 \left[(5 \log 2 - 16 + 6) - (0 - 4 + 3) \right] = 10 \log 2 - 18 \quad \underline{\underline{\text{Ans}}}
 \end{aligned}$$

#Q.

If $F(x) = \int_x^{x^2} \sqrt{\sin t} dt$, then find $F'(x)$

$$F'(x) = \frac{d}{dx} f(x) = \frac{d}{dx} \int_x^{x^2} \sqrt{\sin t} \cdot dt$$

$$= \frac{d}{dx} (x^2) \sqrt{\sin(x^2)} - \frac{d}{dx} (x) \sqrt{\sin(x)}$$

$$= 2x \sqrt{\sin x^2} - \sqrt{\sin x} //$$

[NAT]



#Q.

Evaluate

$$\lim_{x \rightarrow \infty} \frac{\left(\int_0^x e^{t^2} dt \right)^2}{\int_0^x e^{2t^2} dt} = \lim_{x \rightarrow \infty} \frac{\left[\int_0^x e^{t^2} dt \right]^2}{\int_0^x e^{2t^2} dt} = ? = \frac{\infty}{\infty} \text{ form.}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{d}{dx} \left[\int_0^x e^{t^2} dt \right]^2}{\frac{d}{dx} \int_0^x e^{2t^2} dt} = \lim_{x \rightarrow \infty} \frac{2 \left[\int_0^x e^{t^2} dt \right] \cdot \frac{d}{dx} \int_0^x e^{t^2} dt}{\frac{d}{dx} \int_0^x e^{2t^2} dt}$$

$$\lim_{x \rightarrow \infty} \frac{2 \int_0^x e^{t^2} dt \left\{ \frac{d}{dx}(x) e^{x^2} - 0 \right\}}{\left\{ \frac{d}{dx}(x) e^{2x^2} - 0 \right\}} = \lim_{x \rightarrow \infty} \left[\frac{2 \int_0^x e^{t^2} dt}{e^{2x^2}} \right] \approx \frac{\infty}{\infty} = \frac{2[(1) \cdot e^{x^2} - 0]}{e^{2x^2} \cdot (2x)}$$

$$\lim_{x \rightarrow \infty} \left(\frac{1}{x} \right) = 0 \quad \underline{\underline{\text{Ans}}}$$

[MCQ]

#Q. Arc length of the curve $y = x^{3/2}$, $z = 0$ from $(0, 0, 0)$ to $(4, 8, 0)$ is

$$4^{3/2} = (64)^{1/2} = 8$$

$$\frac{dy}{dx} = \frac{3}{2} x^{3/2-1} = \frac{3}{2} \sqrt{x} \Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{9}{4} x$$

$$\text{length} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_0^4 \sqrt{1 + \frac{9}{4}x} dx$$

$$= \frac{1}{2} \int_0^4 (4 + 9x)^{1/2} dx = \frac{1}{2} \left[\frac{(4 + 9x)^{3/2}}{\frac{3}{2} (9)} \right]_0^4 = \frac{1}{27} \left[(40)^{3/2} - 4^{3/2} \right]$$

$$= \frac{8}{27} \left[(10)^{3/2} - 1 \right]$$

A $\frac{8}{27}(10^{3/2} + 1)$

B $\frac{8}{27}(10^{3/2} - 2)$

C $\frac{8}{27}(10^{3/2} - 1)$

D $\frac{8}{27}(10^{3/2} + 2)$

[MCQ]



#Q. Let f be an increasing, differentiable function. If the curve $y = f(x)$ passes through $(1, 1)$ and has length $L = \int_1^2 \sqrt{1 + \frac{1}{4x^2}} dx$, $1 \leq x \leq 2$, then the curve is-

A $y = \ln(\sqrt{x}) - 1$

B $y = 1 - \ln(\sqrt{x})$

C $y = \ln(1 + \sqrt{x})$

D $y = 1 + \ln(\sqrt{x})$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \Rightarrow \frac{dy}{dx} = \frac{1}{2x}$$

$$\int dy = \int \frac{1}{2x} dx + C$$

$$y = \frac{1}{2} \log x + C \Rightarrow y = \log(\sqrt{x}) + 1$$

$$\text{At } (1, 1) \Rightarrow 1 = 0 + C \Rightarrow C = 1$$

#Q. The volume of the solid of revolution of $y = \frac{a}{2}(e^{x/a} + e^{-x/a})$ about x-axis between $x=0$ and $x=b$ is-

- A** $\frac{\pi a^3}{8}(e^{2b/a} - e^{-2b/a}) - \frac{\pi a^2 b}{2}$
- B** $-\frac{\pi a^3}{8}(e^{2b/a} - e^{-2b/a}) + \frac{\pi a^2 b}{2}$
- C** $-\frac{\pi a^3}{8}(e^{2b/a} - e^{-2b/a}) - \frac{\pi a^2 b}{2}$
- D** $\frac{\pi a^3}{8}(e^{2b/a} - e^{-2b/a}) + \frac{\pi a^2 b}{2}$

$$\begin{aligned}
 V &= \int_0^b \pi y^2 dx = \int_0^b \pi \left[\frac{a}{2}(e^{x/a} + e^{-x/a}) \right]^2 dx \\
 &= \frac{\pi a^2}{4} \int_0^b \left(e^{\frac{2x}{a}} + e^{-\frac{2x}{a}} + 2 \right) dx \\
 &= \frac{\pi a^2}{4} \left[\frac{e^{\frac{2x}{a}}}{\left(\frac{2}{a}\right)} + \frac{e^{-\frac{2x}{a}}}{\left(-\frac{2}{a}\right)} + 2x \right]_0^b \\
 &= \frac{\pi a^2}{4} \left[\frac{a}{2} \cdot e^{\frac{2b}{a}} - \frac{a}{2} e^{-\frac{2b}{a}} + 2b \right]
 \end{aligned}$$



THANK - YOU